

ReaguMark: A Resolution-Agnostic Unified Watermarking Framework for Screen-Camera Distortion Resistance

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Abstract—Protecting copyright in Screen-Camera (SC) scenarios is crucial, as content is prone to unauthorized redistribution through physical capture. Watermarking provides an effective solution by embedding imperceptible ownership information in images. Existing SC-resistant watermarking approaches focus mainly on the refinement of differentiable noise layers. However, these methods typically assume fixed input resolutions and struggle to generalize when capture resolutions vary across different scales. This underscores the need for a resolution-agnostic watermarking framework that remains robust under realistic SC degradations. To address these challenges, we propose Resolution-Agnostic Unified Watermarking (ReaguMark), an end-to-end framework for robust watermarking that resists SC distortions and supports arbitrary resolutions. ReaguMark first introduces a resolution-agnostic training strategy that applies multi-scale perturbations and resizes inputs to a unified scale before watermark embedding, improving adaptability to varying resolutions. It further employs an Implicit Neural Representation (INR) to continuously model the watermarked image content, which mitigates discrete resampling aliasing and facilitates watermark embedding across resolutions. In addition, we design a noise layer guided by real SC samples, which comprises components such as Luminance-Adaptive Instance Normalization (LAIN) and Real Moiré Pattern Synthesis (RMPS), facilitating realistic and resolution-consistent degradation simulation. Extensive experiments across various resolutions and real SC conditions demonstrate that ReaguMark can flexibly adapt to arbitrary resolutions without any network reconfiguration or retraining, while still maintaining strong robustness, imperceptibility, and cross-scale consistency.

Index Terms—Robust watermarking, Screen-Camera distortion resistance, resolution-agnostic.

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I. INTRODUCTION

WITH the ubiquity of smartphones and other portable cameras, Screen-Camera (SC) piracy has become an easily accessible yet difficult-to-trace method for unauthorized content distribution [1], [2]. In response, robust watermarking provides a powerful solution for content labeling and accountability tracking.

The primary obstacle for such watermarks lies in the complex degradations inherent to the SC process, including geometric distortion, moiré patterns, and inconsistent illumination, all of which jeopardize the decoding of watermark. Beyond these distortions, a more practical and critical challenge arises from the unpredictable nature of image scale. In real-world scenarios, the resolution (i.e., the scale) of a captured image can vary dramatically due to factors such as screen size, camera distance, and digital zoom. Thus, a watermarking framework tethered to a fixed resolution is impractical for deployment. This makes designing a resolution-agnostic framework that maintains robustness across arbitrary resolutions while resisting SC distortions the significant challenge in watermarking.

Existing watermarking methods have sought to improve robustness in SC scenarios, but most address distortions alone while neglecting resolution variability. On the one hand, most robust SC watermarking schemes prioritize simulating physical distortions to improve robustness [3]–[6]. Although effective against degradations such as moiré and blur, they are fundamentally designed for fixed-resolution images, as depicted in Fig. 1(a). This inflexibility makes them impractical for real-world applications where image scales are unpredictable, as they require costly retraining for each new resolution.

On the other hand, an intuitive attempt at resolution generalization is residual scaling [7], where residuals predicted at a fixed resolution are interpolated to arbitrary sizes. Although simple, this approach is prone to filter distortion and spectral aliasing during discrete interpolation, thereby lacking continuity across scales. Motivated by these limitations, RAIMark [8] stands as the attempt to create a resolution-agnostic framework. However, its complex multi-stage training process, illustrated in Fig. 1(b), achieves cross-scale consistency at the cost of flexibility, rendering the embedded message immutable. More critically, RAIMark overlooks the severe physical distortions of the SC channel, making it robust in the digital domain but fragile in realistic capture scenarios. Consequently, there is

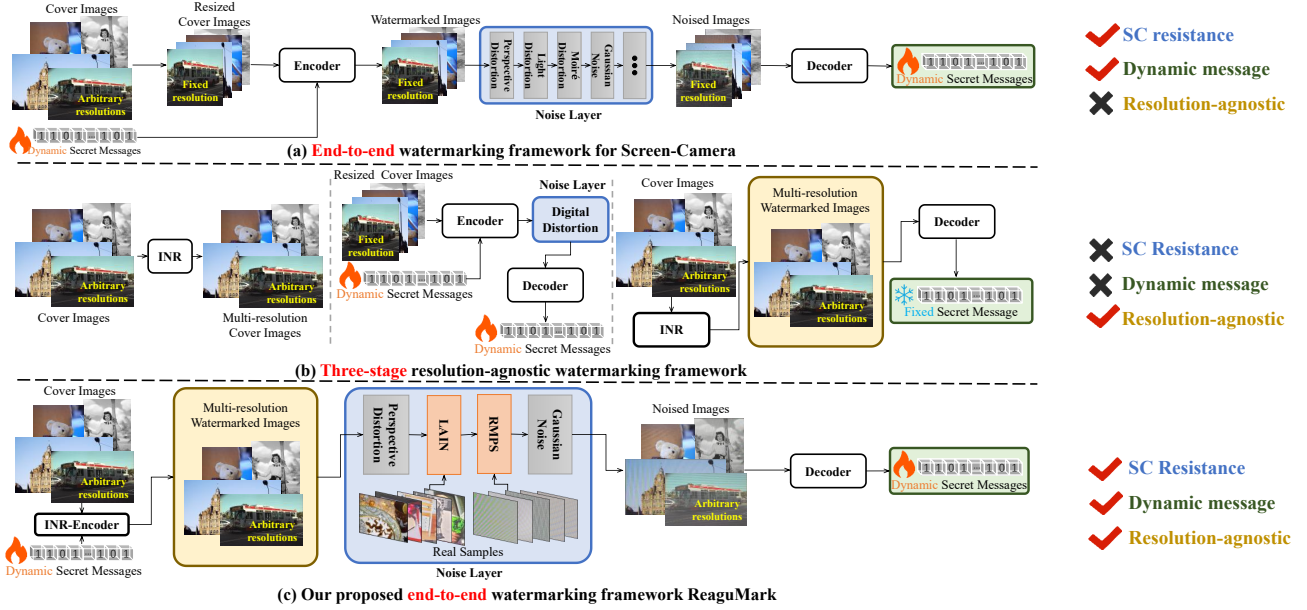


Fig. 1: Comparison of training processes. (a) Conventional fixed-resolution methods require retraining for each new resolution. (b) The three-stage RAIMark framework freezes the decoder, which restricts message flexibility and risks local optima. (c) Our proposed ReaguMark framework employs an end-to-end, resolution-agnostic strategy, leveraging INR for continuous reconstruction while incorporating a real data-guided noise layer to model SC distortions.

a pressing need for a framework that is simultaneously robust against SC distortions and truly resolution-agnostic.

Achieving this dual objective is non-trivial because the robustness against SC degradation is largely determined by the noise simulation strategy used during training. Noise layers play a central role in improving resistance against SC distortions, but existing approaches face distinct limitations.

Model-based noise layers [3], [4], [9], which use mathematical functions for degradations such as moiré and blur, are inherently adaptable to varying resolutions. However, their reliance on simplified noises without incorporating real SC samples prevents them from accurately modeling the distortions encountered in practice, which limits robustness.

In contrast, learning-based noise layers [10], [11] can generate more realistic noise by learning from real data, but their network architectures are typically tied to a specific training resolution. This dependency makes it difficult to generate faithful and consistent degradations across arbitrary resolutions. This predicament highlights the need for a novel noise layer that balances realism, simplicity, and cross-scale adaptability.

To address these challenges, we propose Resolution-Agnostic Unified Watermarking (ReaguMark), an end-to-end framework for robust SC watermarking that supports arbitrary resolutions. As illustrated in Fig. 1(c) and Fig. 2, ReaguMark introduces a resolution-agnostic training strategy that resizes inputs into a unified scale for watermark embedding while applying multi-scale perturbations. Coupled with an Implicit Neural Representation (INR) for continuous content modeling, this design ensures seamless watermark recovery across any resolution without retraining. Furthermore, we design a novel SC noise layer guided by real data. It features a Luminance-

Adaptive Instance Normalization (LAIN) module for color-consistent luminance adaptation and a Real Moiré Pattern Synthesis (RMPS) module to generate realistic, multi-scale moiré pattern. Extensive experiments demonstrate that ReaguMark achieves superior robustness, imperceptibility, and cross-scale consistency especially in SC scenarios.

The contributions of this paper are summarized as follows:

- We propose ReaguMark, an end-to-end screen-camera resistant watermarking framework designed for arbitrary resolutions. It introduces a multi-scale perturbation and unified-scale embedding mechanism, coupled with an Implicit Neural Representation for continuous cross-scale modeling. This design theoretically mitigates the aliasing errors associated with discrete resampling, alleviating the fixed resolution constraint of existing methods and markedly improving adaptability in arbitrary-scale screen-camera scenarios.
- We construct a novel screen-camera noise layer guided by real samples. It integrates a Luminance-Adaptive Instance Normalization module and a Real Moiré Pattern Synthesis module, enabling degradation modeling informed by real screen-camera characteristics with high fidelity and resolution adaptability.
- Extensive experiments conducted under digital and screen-camera conditions demonstrate that our method outperforms state-of-the-art schemes in terms of robustness, imperceptibility, and cross-scale consistency.

Section II reviews related work on watermarking methods. Section III details the proposed method, including framework and loss function design. Section IV presents experimental results, demonstrating the performance of our method. Finally, Section V concludes our work and outlines future directions.

II. RELATED WORK

In this section, we review prior work on robust watermarking, with a focus on deep learning-based, SC resilient, and resolution-agnostic watermarking methods.

A. Robust Watermarking based on Deep Learning

Traditional robust watermarking techniques operate in the spatial domain [12] by directly modifying pixel values, or in transform domains such as the Discrete Cosine Transform (DCT) [13], the Discrete Wavelet Transform (DWT) [14], and Singular Value Decomposition (SVD) [15] to achieve better robustness. However, the performance of these hand-crafted methods remains limited.

With the advent of deep learning, the field has been significantly advanced [16]. HiDDeN [17] pioneered the end-to-end framework, jointly training an encoder, decoder, and a differentiable noise layer to optimize both imperceptibility and robustness. Subsequent works have introduced various architectural and training refinements. For instance, TSDDL [18] proposed a two-stage training scheme to handle non-differentiable distortions, while MBRS [19] specifically targeted JPEG compression by incorporating a differentiable simulator. Recently, WAM [20] improved resistance to local edits by extracting messages from multiple image patches.

As task complexity grows, multi-stage training has become a common strategy to stabilize the optimization process, applied in scenarios like adaptive embedding strength [21] and geometric distortion resistance [22]. However, this approach often lacks a unified global objective, which can lead to suboptimal solutions due to the lack of synergy between stages [23]. Our ReaguMark framework returns to a fully end-to-end pipeline to overcome this limitation.

B. Screen-Camera Resilient Watermarking

The SC channel presents a unique and formidable challenge due to its complex physical distortions. Early efforts employed traditional watermarking techniques using the template [24], [25]. Subsequently, deep learning methods [3], [4], [26] emerged as the predominant approach, where the design of the differentiable noise layer is paramount. These noise layers generally follow two distinct strategies: model-based simulation and data-driven learning.

Model-based approaches rely on mathematical functions to simulate key distortions. The pioneering work, StegaStamp [3], introduced a noise layer simulating blur, color shifts, JPEG compression, and other distortions to prepare the model for physical-world scenarios. Building on this, PIMoG [4] identified perspective, illumination, and moiré patterns as the most critical SC distortion, demonstrating that focusing on core distortions yields significant gains. Other works used template synchronization [27] or homography restoration [28] to improve the robustness. However, the lack of guidance from real SC samples limits the ability of model-based methods to capture practical distortions, constraining their robustness.

To improve the realism of distortion simulation, data-driven approaches have emerged. For instance, CDTF [10]

first learned the SC transfer function directly from the data. More recently, SRWSS [11] employed a style transfer network trained on real SC images to generate a more faithful noise layer. Although these methods achieve strong robustness through realistic simulation, their underlying network architectures are typically tied to a fixed training resolution. This dependency makes it difficult to generate consistent degradations across arbitrary scales, limiting their practical adaptability. Our work addresses this dilemma by designing a noise layer that is both real-data-guided and resolution-agnostic.

C. Resolution-Agnostic Watermarking

A long-standing goal in image processing is to develop models that can operate at arbitrary resolutions without retraining. In watermarking, this capability is crucial for practical deployment, but has been rarely explored. A straightforward strategy is residual scaling [7], where the encoder is trained at a fixed resolution and the predicted residual is interpolated to arbitrary sizes during inference. Although this technique avoids retraining and extends applicability to diverse input resolutions, it essentially performs discrete resampling of residuals and thus fails to achieve true continuity across scales. As a result, it often suffers from information loss and inconsistency when facing large resolution shifts.

To our knowledge, RAIMark [8] was the first to explicitly tackle resolution-agnostic watermarking by leveraging Implicit Neural Representation (INR). It embeds a message by fine-tuning a pre-fitted INR of a cover image, enabling watermark extraction at any scale. This design provides a more principled way of handling arbitrary resolutions compared to residual scaling. However, RAIMark suffers from two critical limitations. First, its three-stage training process, which involves pre-fitting an INR, pre-training a decoder, and fine-tuning the INR, is not a unified optimization. A direct consequence is the freezing of the decoder in the final stage, which makes the embedded message immutable and reduces flexibility. Second, and more importantly, for SC scenarios, the RAIMark noise layer only accounts for digital distortions, leaving it vulnerable to the physical degradations of the SC channel.

Our proposed ReaguMark framework is designed to overcome both limitations. By integrating INR into an end-to-end architecture and introducing a real-data-guided SC noise layer, it achieves the first watermarking solution that is simultaneously resolution-agnostic and robust to SC distortions.

III. PROPOSED METHOD

In this section, we provide an overview of the framework of the watermarking, followed by a detailed description of each component.

A. Method Overview

The proposed ReaguMark framework is illustrated in Fig. 2. The overall architecture consists of three main modules: Encoding, Noise Addition, and Decoding, supporting resolution-agnostic watermarking throughout the entire pipeline. In Encoding, the encoder receives an cover image I_{co} of arbitrary

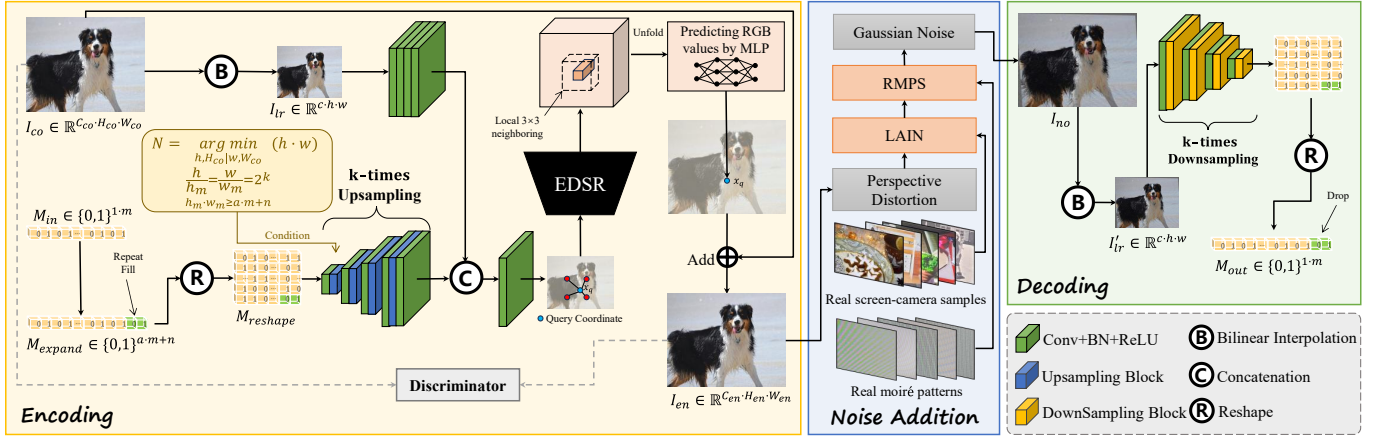


Fig. 2: The overall architecture of the proposed ReaguMark framework. The end-to-end pipeline supports arbitrary resolutions. The Encoding module uses a resolution-agnostic strategy and an INR-based generator to embed a message into a cover image. The Noise Addition module simulates realistic SC degradations guided by real samples. The Decoding module extracts the message from the distorted image at any resolution.

resolutions and a message M_{in} to embed, producing a watermarked image I_{en} . The encoder accommodates different input and output resolutions, supporting flexible resizing requirements. In Noise Addition, I_{en} passes through the SC noise layer guided by real degradation samples to mimic SC interferences, enhancing robustness under complex degradations. This stage outputs I_{no} . In Decoding, the decoder extracts the embedded message M_{out} from I_{no} of any resolution, ensuring stable and accurate recovery across scales.

B. Encoding

For encoding, I_{co} is first randomly scaled and resized on a standard scale to extract main features. Meanwhile, the message is reshaped into $M_{reshape}$ through replication and filling, concatenated with the image features, and fused via convolution to form a unified representation. Finally, an INR-based network generates I_{en} at any resolution. A discriminator is also used to evaluate image realism and improve embedding quality.

1) Multi-Scale Perturbation and Unified-Scale Encoding:

In the natural propagation of images, resolution changes are common. If a model is trained directly on a fixed size, the watermark embedding may fail in scenarios involving scaling. To enhance the model's scale generalization ability, this work first introduces a multi-scale scale perturbation: given $I_{co} \in \mathbb{R}^{C_{co} \times H_{co} \times W_{co}}$, a scale factor $s \sim U(s_{min}, s_{max})$ is sampled from a set range. The image is resized to (H_{co}, W_{co}) , where $(H_{co}, W_{co}) = (h \cdot s, w \cdot s)$, and then resized to a standard size (h, w) before input to the encoder. The unified-scale encoding strategy keeps the network architecture simple at the standard scale, avoiding the need for multi-scale feature fusion modules, thus reducing implementation complexity while improving training stability and efficiency.

2) *Structured Spatial Mapping of Low-Dimensional Message:* To embed a message into the image feature space, the binary input $M_{in} \in \{0, 1\}^{1 \times m}$ must be reshaped into a spatially distributed feature aligned with the image features.

We design a message processing module that first expands M_{in} by redundant replication and zero-padding to length $a \cdot m + n$, then reshapes it into a feature map of dimension (h_m, w_m) , ensuring spatial alignment with the image feature map. The choice of (h_m, w_m) must satisfy the following:

$$\frac{h}{h_m} = \frac{w}{w_m} = 2^k \quad \text{and} \quad h_m \cdot w_m \geq a \cdot m + n, \quad (1)$$

where a is a redundancy factor, representing the number of times the original message m is replicated and n is the number of message bits used to fill the remaining space in the feature map of size (h_m, w_m) . So that the feature can be upsampled k times to the image scale, constructing a stable and general spatial mapping mechanism. The upsampling consists of transposed convolution, Batch Normalization, and ReLU. This module does not require architectural changes for specific input sizes or message lengths and uniformly injects information into images of arbitrary resolutions.

3) *Feature Fusion and Unified Embedding Representation:* The processed image and message features are concatenated at a unified scale and fused through multiple convolutional layers into a single representation. This fused feature encodes both image and message features, serving as an intermediate representation that enables the effective embedding of the message into the image.

4) *Arbitrary-Resolution Reconstruction based on INR:* To achieve resolution-agnostic watermark generation, we employ an arbitrary-resolution reconstruction mechanism based on Implicit Neural Representation (INR). INR has emerged as a powerful paradigm for representing signals as continuous functions learned by a neural network [29]. Instead of representing an image as a discrete grid of pixels, an INR-based model learns a mapping from spatial coordinates to their corresponding signal values (e.g., RGB colors). This continuous representation allows the image to be sampled and rendered at any desired resolution. Specifically, the model does not directly output a fixed-resolution image. Instead, it dynamically generates the RGB value for any given coordinate

point by querying, enabling watermark recovery at any resolution. In terms of network architecture, we first use a feature extractor such as EDSR [30] to extract deep feature maps of fixed resolution from the input. Subsequently, for any given resolution, the pixel coordinate x_q , its corresponding local region feature F_x , the relative coordinate $x_q - v_{x_q}$, frequency positional encoding $\gamma(x_q)$, and the cell scale information $cell$ are concatenated as input. A Multi Layer Perceptron (MLP) then implements the mapping function from coordinates to RGB values:

$$I(x_q) = f_\theta(z_x), \quad (2)$$

where $z_x = \text{Concat}(F_x, x_q - v_{x_q}, \gamma(x_q), cell)$, $I(x_q)$ is the RGB value generated at position x_q , and θ represents the network parameters. Finally, a local integration strategy [31] is employed: four offset samples are taken around the query point, and a weighted region is constructed based on relative coordinates. The final output is an area-weighted fusion of the predictions from each direction, enhancing the continuity of the prediction.

Simultaneously, to better fit high-frequency variations, we incorporate frequency positional encoding into the input, following [32], represented as:

$$\gamma(x) = (\sin(2\pi \cdot 2^0 \cdot x), \cos(2\pi \cdot 2^0 \cdot x), \dots, \sin(2\pi \cdot 2^{L-1} \cdot x), \cos(2\pi \cdot 2^{L-1} \cdot x)), \quad (3)$$

where L is a hyperparameter that determines the dimensionality, set to 4. By concatenating the positional encoding with the MLP input, the network can more easily capture texture details and edge variations, ultimately achieving high-quality, cross-scale watermarked image generation.

5) *Theoretical Analysis of Resolution-Agnostic Representation*: Achieving consistent watermark embedding across arbitrary resolutions requires the representation to be invariant to scaling. Specifically, changing the output resolution should not introduce additional distortion to the watermark signal. However, in order to support arbitrary resolutions, conventional deep watermarking methods commonly rely on a discrete residual scaling strategy [7], which introduces resolution-dependent degradation under practical conditions.

To theoretically characterize this limitation, we analyze the fundamental mechanism of this widely adopted scaling approach. The process can be abstracted as obtaining a discrete sampled representation of a continuous watermark signal $w(x, y)$ on a low-resolution grid, followed by interpolation and resampling to higher resolutions. To analyze this process, we rely on the continuous-discrete image resampling framework formalized by Thévenaz et al. [33]. Let $w(x, y)$ denote an idealized continuous signal model of the watermark, and $W(u, v)$ its Fourier transform, where (u, v) represents the continuous frequency coordinates. Let Δ_{lr} and Δ_{hr} denote the sampling intervals of the low- and high-resolution grids, respectively, with $\Delta_{hr} < \Delta_{lr}$.

The discrete residual $w_s(x, y)$ corresponds to sampling $w(x, y)$ on a lattice with spacing Δ_{lr} , written as:

$$w_s(x, y) = w(x, y) \cdot \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} \delta(x - i\Delta_{lr}, y - j\Delta_{lr}), \quad (4)$$

where $\delta(\cdot)$ denotes the Dirac delta function, and $i, j \in \mathbb{Z}$ are the lattice sampling indices.

Its spectrum, $W_s(u, v)$, is given by periodic replication in the frequency domain:

$$W_s(u, v) = \frac{1}{\Delta_{lr}^2} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} W\left(u - \frac{k}{\Delta_{lr}}, v - \frac{l}{\Delta_{lr}}\right), \quad (5)$$

where $k, l \in \mathbb{Z}$ represent the spectral replication indices.

To generate higher-resolution outputs, an interpolation operator with kernel $h(x, y)$ is applied to obtain a continuous reconstruction:

$$\hat{w}(x, y) = w_s(x, y) * h(x, y), \quad (6)$$

with frequency-domain representation:

$$\hat{W}(u, v) = \frac{H(u, v)}{\Delta_{lr}^2} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} W\left(u - \frac{k}{\Delta_{lr}}, v - \frac{l}{\Delta_{lr}}\right), \quad (7)$$

where $H(u, v)$ is the frequency response of $h(x, y)$. The final high-resolution signal is obtained by sampling $\hat{w}(x, y)$ on the grid with interval Δ_{hr} :

$$\hat{w}_{hr}[m, n] = \hat{w}(m\Delta_{hr}, n\Delta_{hr}). \quad (8)$$

Before the final resampling step, the continuous reconstruction error $E(u, v) = W(u, v) - \hat{W}(u, v)$ can be decomposed as:

$$E(u, v) = \underbrace{W(u, v) \left[1 - \frac{H(u, v)}{\Delta_{lr}^2}\right]}_{\text{Filter Distortion}} - \underbrace{\frac{H(u, v)}{\Delta_{lr}^2} \sum_{(k, l) \neq (0, 0)} W\left(u - \frac{k}{\Delta_{lr}}, v - \frac{l}{\Delta_{lr}}\right)}_{\text{Aliasing Error}}, \quad (9)$$

where the $(0, 0)$ term corresponds to the baseband component, and the remaining terms represent spectral aliasing. Note that the frequency-domain error $E(u, v)$ corresponds to spatial-domain distortion via the Fourier transform, and thus directly affects the reconstructed signal quality. Furthermore, the final discrete output $\hat{w}_{hr}[m, n]$ is obtained by sampling $\hat{w}(x, y)$. Consequently, the total error of the traditional discrete pipeline inherently encompasses this continuous reconstruction error $E(u, v)$, which propagates to the discrete domain through subsequent sampling and directly affects the observed pixel-level distortion, compounded by additional aliasing if $\hat{w}(x, y)$ is not strictly band-limited to the target Nyquist frequency $\frac{1}{2\Delta_{hr}}$.

Perfect reconstruction ($E(u, v) = 0$ and no subsequent resampling aliasing) holds if and only if two conditions are simultaneously satisfied: (i) $H(u, v)$ corresponds to an ideal low-pass filter with $H(u, v) = \Delta_{lr}^2$ for $|u|, |v| < \frac{1}{2\Delta_{lr}}$ and zero elsewhere, and (ii) $W(u, v) = 0$ for $|u|, |v| \geq \frac{1}{2\Delta_{lr}}$. These conditions correspond to the Nyquist sampling criterion combined with ideal reconstruction filtering. As analyzed by Unser [34], these conditions are generally not satisfied in practical settings. Practical watermark signals are rarely strictly band-limited, which inherently introduces sampling

aliasing. Furthermore, interpolation kernels deviate from ideal filters, causing unavoidable passband attenuation and failing to suppress spectral replicas. Consequently, the reconstruction error is non-zero under practical conditions, indicating that discrete resampling introduces scale-dependent distortion.

To overcome this limitation, we leverage the continuous image representation principles introduced by Chen et al. [31] and model the watermark via a continuous mapping $\Phi_\theta : \Omega \rightarrow \mathbb{R}^C$ over the normalized coordinate domain $\Omega = [-1, 1]^2$ to learn a continuous function that approximates the signal. Here, $x_q = (x, y)$ denotes the 2D continuous coordinate, Φ_θ represents the effective continuous mapping parameterized by our MLP f_θ , and $w(x_q)$ designates the idealized signal value evaluated at x_q . The watermark at an arbitrary target resolution is obtained via direct coordinate querying:

$$w_{\text{LIIF}}[m, n] = \Phi_\theta(\mathcal{M}_{hr}(m, n)), \quad (10)$$

where $m, n \in \mathbb{Z}$ denote the discrete pixel indices in the target high-resolution grid, and $\mathcal{M}_{hr} : \mathbb{Z}^2 \rightarrow \Omega$ maps these indices to continuous coordinates x_q .

The generation error is characterized by the approximation residual $\epsilon(x_q)$:

$$\epsilon(x_q) = w(x_q) - \Phi_\theta(x_q), \quad (11)$$

where $w(x_q)$ denotes the idealized continuous signal. In contrast to discrete resampling, which reconstructs an approximation $\hat{w}(x, y)$ from sampled observations, the proposed method learns a continuous function that approximates the signal. Thus, for any queried coordinate $x_q = \mathcal{M}_{hr}(m, n)$, the effective reconstruction error $E_{\text{INR}}(x_q)$ satisfies:

$$E_{\text{INR}}(x_q) = \epsilon(x_q). \quad (12)$$

This error is primarily determined by the representation and approximation capability of the MLP f_θ and does not introduce the resolution-ratio-dependent aliasing terms that arise from explicit discrete reconstruction operators. Therefore, scaling is realized as a change in querying density over a shared continuous representation Φ_θ . Because this process directly samples from the learned continuous mapping rather than reconstructing an intermediate signal through fixed discrete kernels, the dominant scale-dependent error mechanism of discrete residual resampling is absent by construction. Notably, under an ideal condition where the network approximation error is zero, this continuous formulation theoretically minimizes scaling error, whereas the traditional discrete pipeline inherently retains filter distortion and aliasing errors. This approach therefore offers lower approximation error for scale-invariant watermark representation compared to the discrete residual pipeline, while the remaining error is primarily governed by the approximation capability of the continuous representation Φ_θ .

6) *Analysis of the Coordinate Based MLP*: Given that the continuous generation of watermarks inherently depends on the expressive power of the network, we further analyzed the coordinate-based MLP from the perspectives of representational capacity and computational complexity to provide a basis for our architectural configuration.

a) *Representational Capacity*: Standard coordinate-based MLPs inherently struggle to represent natural and complex images with high fidelity [35]. They typically suffer from spectral bias, prioritizing low-frequency functions and failing to capture high-frequency variations. Since imperceptible watermarking requires modulation of the host image's complex textures across various frequency bands, a naive MLP lacks the required representational capacity. To overcome this limitation, our framework leverages the representational paradigm introduced by the Local Implicit Image Function (LIIF) [31]. In LIIF, an image is represented as a set of latent codes distributed in spatial dimensions. The decoding MLP predicts the signal by taking the coordinate information and querying the local latent codes around the coordinate as inputs. By conditioning our MLP on the local deep features extracted from the cover image, our framework inherits this enhanced representational capacity to recover fine details.

b) *Computational Complexity*: Unlike Convolutional Neural Networks, where the memory and computational costs scale linearly with the total number of pixels $\mathcal{O}(H \times W)$, the complexity of our INR generator depends purely on the number of sampled coordinate queries, denoted as $\mathcal{O}(N_{\text{query}})$. Although generating high-resolution images demands substantial memory allocation, the point-wise nature of INR allows coordinate queries to be executed in independent batches, effectively mitigating GPU memory constraints during inference. Consequently, this design allows ReaguMark to efficiently learn the continuous watermarking representation during training and flexibly apply it across various scales during inference.

C. Noise Addition

The complete computational pipeline for the SC degradation simulation, including perspective transform, statistical matching in LAIN, and blending mechanisms in RMPS, is detailed in Algorithm 1.

1) *Perspective Distortion*: To enhance the model's robustness against angular changes and geometric distortions in SC scenarios, we introduce a perspective transformation mechanism. By applying random offsets to the four corner points of the image, we construct a mapping between the source and target points and compute the perspective transformation matrix M . This allows for mapping from the cover image to the distorted image:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = M \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad (13)$$

where M is a random homography matrix, $(x, y, 1)$ are the coordinates in the cover image and (x', y', z') are the transformed coordinates. This perturbation effectively simulates the common perspective shifts encountered in SC, helping to improve the robustness of the watermark embedding and extraction process. [36]

2) *Luminance-Adaptive Instance Normalization (LAIN)*: To achieve more controllable and content-preserving adjustment of image luminance, we designed a Luminance-Adaptive Instance Normalization module. As shown in Fig. 3(a), this

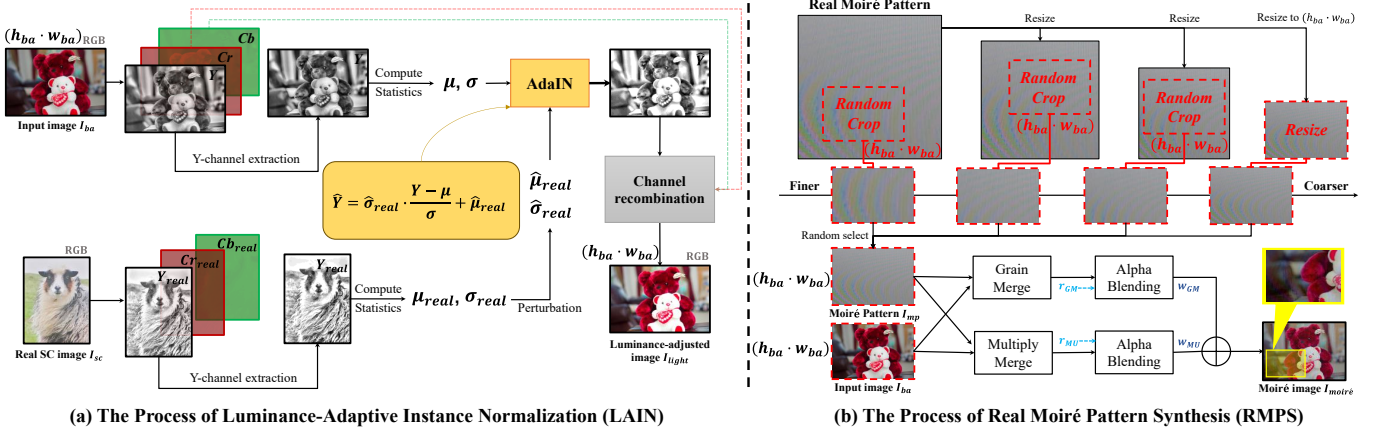


Fig. 3: The process of the proposed (a) Luminance-Adaptive Instance Normalization (LAIN) and (b) Real Moiré Pattern Synthesis (RMPS). In LAIN, image luminance is normalized based on statistics from a real SC sample. In RMPS, a real moiré pattern is pre-processed and blended with the image using dual strategies to generate realistic and structurally diverse interference.

Algorithm 1 Screen Camera Degradation Simulation

- 1: **Input:** Encoded image $I_{en} \in \mathbb{R}^{3 \times H \times W}$, Real SC image I_{sc} , Real Moiré pattern M_{real}
- 2: **Output:** Distorted image I_{no}
- 3: **Step 1: Perspective Transform**
- 4: $I_{persp} \leftarrow \text{PerspectiveTransform}(I_{en}, \text{distance})$
- 5: **Step 2: Luminance Adaptive Instance Normalization**
- 6: $Y_{en}, Cb_{en}, Cr_{en} \leftarrow \text{RGB2YCbCr}(I_{persp})$
- 7: $Y_{sc} \leftarrow \text{RGB2YCbCr}(I_{sc})$
- 8: $\mu_{en}, \sigma_{en} \leftarrow \text{MeanAndStd}(Y_{en})$
- 9: $\mu_{sc}, \sigma_{sc} \leftarrow \text{MeanAndStd}(Y_{sc})$
- 10: $\hat{\mu}_{sc} \leftarrow \mu_{sc} \cdot (1 + \epsilon_{\mu})$
- 11: $\hat{\sigma}_{sc} \leftarrow \sigma_{sc} \cdot (1 + \epsilon_{\sigma})$
- 12: $\hat{Y}_{en} \leftarrow \hat{\sigma}_{sc} \frac{Y_{en} - \mu_{en}}{\sigma_{en}} + \hat{\mu}_{sc}$
- 13: $I_{light} \leftarrow \text{YCbCr2RGB}(\hat{Y}_{en}, Cb_{en}, Cr_{en})$
- 14: **Step 3: Real Moiré Pattern Synthesis**
- 15: $I_{ba} \leftarrow I_{light}$
- 16: $I_{mp} \leftarrow \text{RandomCropAndResize}(M_{real}, (h_{ba}, w_{ba}))$
- 17: $I_{sm}^{MU} \leftarrow I_{ba} \odot I_{mp}$
- 18: $I_{sm}^{GM} \leftarrow I_{ba} + I_{mp} - 0.5$
- 19: $I_{comp}^{MU} \leftarrow \text{AlphaBlending}(I_{ba}, I_{sm}^{MU}, \text{op}_{MU})$
- 20: $I_{comp}^{GM} \leftarrow \text{AlphaBlending}(I_{ba}, I_{sm}^{GM}, \text{op}_{GM})$
- 21: $I_{moiré} \leftarrow w_{MU} \cdot I_{comp}^{MU} + w_{GM} \cdot I_{comp}^{GM}$
- 22: **Step 4: Final Processing**
- 23: $I_{no} \leftarrow I_{moiré} + \text{GaussianNoise}(\sigma)$
- 24: **return** I_{no}

module operates in the YCbCr color space, performing normalization only on the luminance channel (Y) and introducing a controllable perturbation mechanism to enhance the diversity of SC lighting conditions:

a) *Color Space Conversion:* The input image I_{ba} and a randomly sampled real SC image I_{sc} (taken from the SC dataset in [36]) are first converted from RGB to YCbCr space. We only select the luminance channel (Y) for normalization to avoid color distortion.

b) *Feature Statistics Calculation:* For the luminance channels of I_{ba} and I_{sc} , we compute their mean and standard deviation along the channel dimension, which describe the luminance distribution characteristics of the images.

c) *Perturbation Mechanism:* We introduce perturbation factors for the mean and standard deviation to simulate subtle luminance variations, enhancing the diversity and unpredictability of the luminance transfer results. Let the mean and standard deviation of the Y channel of I_{sc} be μ_{real} and σ_{real} , and the perturbation factors be $\epsilon_{\mu} \in [-0.1, 0.1]$ and $\epsilon_{\sigma} \in [-0.05, 0.05]$. The perturbed statistics are then:

$$\hat{\mu}_{real} = \mu_{real} \cdot (1 + \epsilon_{\mu}), \quad (14)$$

$$\hat{\sigma}_{real} = \sigma_{real} \cdot (1 + \epsilon_{\sigma}). \quad (15)$$

d) *AdaIN Operation:* Based on AdaIN [37], the Y channel of I_{ba} is normalized and then remapped to the perturbed statistics of I_{sc} :

$$\hat{Y} = \hat{\sigma}_{real} \left(\frac{Y_{en} - \mu_{en}}{\sigma_{en}} \right) + \hat{\mu}_{real}. \quad (16)$$

e) *Channel Recombination and Inverse Color Space Transform:* The chrominance channels (Cb, Cr) of the content image remain unchanged. The modified luminance channel is recombined with the original chrominance channels, and the image is converted back from YCbCr to the RGB space to obtain the final image I_{light} .

3) *Real Moiré Pattern Synthesis (RMPS):* Traditional moiré pattern synthesis methods [38] based on mathematical models struggle to capture the complex structures of real-world moiré patterns. On the other hand, end-to-end synthesis networks are computationally expensive and lack adaptability to multiple resolutions. To overcome these challenges, as shown in Fig. 3(b), we adopt a more direct and efficient method: overlaying pure moiré patterns extracted from real captures as noise onto clean images.

a) *Moiré Pattern Processing:* Specifically, we utilize the publicly available UniDemoiré dataset [39] (<https://github.com/UniDemoiré/UniDemoiré>).

[//github.com/4DVLab/UniDemoire](https://github.com/4DVLab/UniDemoire)), which contains high-quality moiré patterns captured on a clean white background. To enhance spatial adaptability, we design a randomized multi-scale cropping and scaling strategy. Given a target image tensor, we first randomly crop a random patch from the high-resolution moiré source image, which is subsequently resized to match the exact spatial dimensions of the target tensor. This dynamic sampling provides a diverse set of moiré patterns for the subsequent fusion stage.

b) Moiré Pattern Blending: To synthesize more realistic moiré-contaminated images, we follow the blending strategies described in the UniDemoire work, combining two complementary image blending strategies: Multiply (MU) [40] and Grain Merge (GM) [41]. The former simulates destructive interference effects through intensity modulation, while the latter enhances the image through brightness superposition, capturing aliasing features from the SC process.

Let $I_{ba} \in \mathbb{R}^{C \times H \times W}$ be the input background image. To obtain the moiré overlays, we first randomly sample a high-resolution real moiré image M_{real} from UniDemoire dataset. Crucially, because the processed moiré patterns I_{mp} originate strictly from blank white screen captures in the dataset, they contain zero semantic content. Therefore, these blending operations effectively ensure that only pure optical interference frequencies are transferred, effectively avoiding any target image information from leaking into the moiré overlays during the synthesis process. We then apply random cropping and resizing to generate the processed moiré pattern I_{mp} . We first use the multiply strategy $MU(\cdot, \cdot)$ for synthesis:

$$I_{sm}^{MU} = MU(I_{ba}, I_{mp}) = I_{ba} \odot I_{mp}, \quad (17)$$

where $\odot$ denotes element-wise multiplication.

Simultaneously, we use GM for synthesis, a brighter strategy that can balance the darker results of $MU(\cdot, \cdot)$:

$$I_{sm}^{GM} = GM(I_{ba}, I_{mp}) = I_{ba} + I_{mp} - 0.5. \quad (18)$$

Subsequently, we employ Alpha Blending [42] for opacity-aware image synthesis to simulate varying degrees of layer transparency:

$$I_{comp}^{MU} = r_{MU} \cdot MU(I_{ba}, I_{mp}) + (1 - r_{MU}) \cdot I_{ba}, \quad (19)$$

$$I_{comp}^{GM} = r_{GM} \cdot GM(I_{ba}, I_{mp}) + (1 - r_{GM}) \cdot I_{ba}, \quad (20)$$

where r_{MU} and r_{GM} represent the synthesis weights of the MU and GM output layers, respectively. Their calculation is based on the opacity relationship between the foreground and background layers:

$$r_x = \frac{op_x}{op_x + (1 - op_x) \cdot op_{ba}}, \quad x \in \{MU, GM\}. \quad (21)$$

Here, op_{MU} and op_{GM} are the opacities of the MU and GM, and op_{ba} is the opacity of the background image. Finally, we linearly combine the two synthesis results with adjustable weights to obtain the simulated moiré image:

$$I_{moiré} = w_{MU} \cdot I_{comp}^{MU} + w_{GM} \cdot I_{comp}^{GM}, \quad (22)$$

where the weights w_{MU} and w_{GM} are independently adjusted to control the brightness and moiré intensity of the fused result,

set to 0.125 and 0.1. This dual-fusion strategy effectively enhances the structural diversity and controllability of the moiré simulation process, resulting in generated images that are visually closer to real SC scenarios.

Finally, following the recommendation in PIMoG [4], we add Gaussian noise.

D. Decoding

The decoding module mirrors the encoding process in reverse. The input watermarked image, regardless of resolution, is first rescaled to the unified size (h, w) , then passed through a convolutional network symmetrical to the encoder to obtain a feature map of size (h_m, w_m) . This map is reshaped into a vector, from which the original m -bit message is recovered by selecting the corresponding positions from the expanded representation. This unified-scale decoding allows robust message extraction under arbitrary resolutions in SC scenarios.

The primary goal of the decoding phase is to accurately retrieve the embedded message regardless of the input resolution. The decoder accepts an input image of any resolution and rescales it to a fixed canonical resolution (h, w) for extraction, making the decoding process equally resolution-agnostic as the encoding phase. This design completes the end-to-end resolution-agnostic pipeline: the encoder embeds the message into images of arbitrary resolution via INR-based continuous generation, and the decoder extracts the message from images of arbitrary resolution via resolution-normalized extraction. The embedded message can therefore be reliably retrieved across diverse input resolutions without any resolution-specific adaptation at either end of the pipeline.

E. Loss Functions

1) Image Quality Loss: To ensure that the watermarked image I_{en} and the cover image I_{co} maintain high visual quality, we introduce a combination of multiple image similarity losses for joint optimization:

a) Pixel-wise Reconstruction Loss: Minimizes the Mean Squared Error (MSE) between the encoded image and the cover image in pixel space: $\mathcal{L}_P = \|I_{co} - I_{en}\|_2^2$.

b) Structural Similarity Loss: Captures the Multi-Scale Structural Similarity Index Measure (MS-SSIM) of structural information in images, enhancing the perceptual quality of the encoded image [43]: $\mathcal{L}_S = 1 - \text{MS-SSIM}(I_{co}, I_{en})$.

c) Perceptual Loss: Introduces a distance metric based on deep network perceptual features to measure high-level semantic consistency [44]: $\mathcal{L}_L = \text{LPIPS}(I_{co}, I_{en})$.

The final image quality loss is a weighted sum of the three terms:

$$\mathcal{L}_{en} = \lambda_P \mathcal{L}_P + \lambda_S \mathcal{L}_S + \lambda_L \mathcal{L}_L, \quad (23)$$

where λ_P , λ_S , and λ_L are weight hyperparameters.

2) Decoding Loss: To ensure that embedded messages can be accurately recovered after noise interference, we use the MSE between the decoded message M_{out} and the original message M_{in} : $\mathcal{L}_M = \|M_{in} - M_{out}\|_2^2$. The decoding loss is: $\mathcal{L}_{de} = \lambda_M \mathcal{L}_M$.

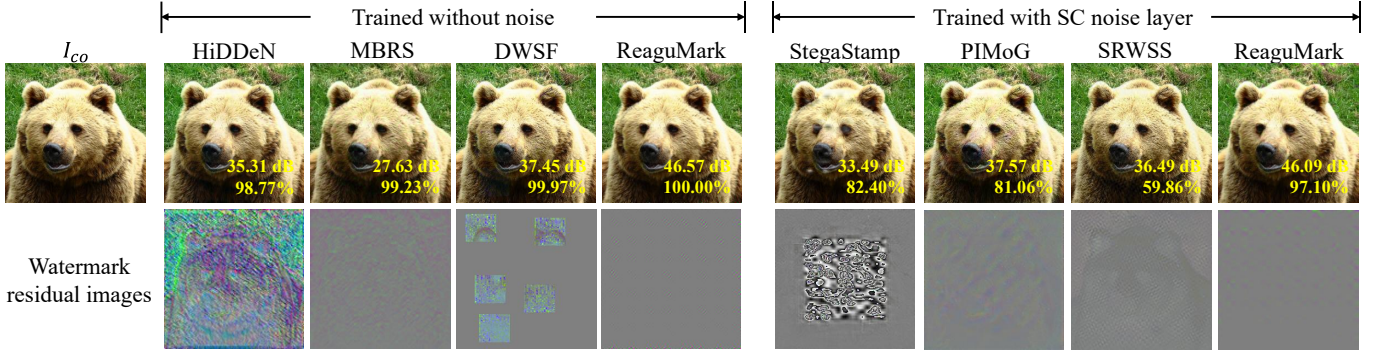


Fig. 4: Visualization of watermarked images generated by different methods. Our ReaguMark produces images that are visually similar to the cover, demonstrating great imperceptibility.

TABLE I: Digital noise robustness comparison. We report ACC (%) for robustness and PSNR (dB) for imperceptibility. Our method achieves state-of-the-art robustness with significantly higher visual quality. The model used for JPEG is trained by JpegSS [17]. † indicates metrics directly cited from the original paper.

Models	Invisibility	Robustness (%)							AVG
	PSNR (dB)	Identity	GN(0.05)	MF(7)	JPEG(50)	Crop(0.25)	Resize(0.5)	Resize(2.0)	
HiDDeN	35.31	98.77	94.80	94.80	60.77	98.53	98.67	99.13	92.21
MBRS	27.63	99.23	97.90	98.97	97.43	58.80	98.97	99.27	92.94
DWSF	37.45	99.97	99.97	100.00	95.83	51.33	99.97	100.00	92.44
RAIMark†	39.61	100.00	99.97	100.00	99.97	99.20	100.00	100.00	99.88
ReaguMark	46.57	100.00	100.00	100.00	99.26	99.95	100.00	100.00	99.89

3) *Adversarial Loss*: To enhance the realism of the generated images, a discriminator network is introduced for adversarial training. Specifically, the discriminator learns to distinguish between cover and watermarked images, while the encoder aims to fool the discriminator.

a) *Discriminator Loss*: $\mathcal{L}_D = \mathcal{L}_{BCE}(D(I_{co}), 1) + \mathcal{L}_{BCE}(D(I_{en}), 0)$.

b) *Encoder Loss*: $\mathcal{L}_{adv} = \mathcal{L}_{BCE}(D(I_{en}), 1)$.

4) *Total Loss*: Considering the three types of losses, the final joint optimization objective for the encoder and decoder is the following:

$$\mathcal{L}_{total} = \lambda_{en}\mathcal{L}_{en} + \lambda_{de}\mathcal{L}_{de} + \lambda_{adv}\mathcal{L}_{adv}, \quad (24)$$

where λ_{en} , λ_{de} , and λ_{adv} are weight hyperparameters. The discriminator loss $\mathcal{L}_D$ is used exclusively for training the discriminator and does not participate in the optimization of the encoder and decoder. All parameters are selected following commonly adopted settings in prior watermarking and super-resolution studies, and further adjusted to our network structure and training objectives. The final settings in our experiments are $\lambda_P = 1.0$, $\lambda_S = 1.0$, $\lambda_L = 5.0$, $\lambda_M = 10.0$, $\lambda_{en} = 1.0$, $\lambda_{de} = 1.0$, and $\lambda_{adv} = 0.0001$.

IV. EXPERIMENTS

In this section, we evaluate ReaguMark through experiments on imperceptibility, robustness to digital and SC distortions, and cross-scale consistency.

A. Implementation Details

To ensure a fair comparison and maintain consistency with prior evaluations, we adopt the COCO dataset [45] used in

SRWSS. Specifically, we randomly sample 10,000 images for training and a disjoint set of 3,000 images for testing. For each image, a randomly generated 30-bit string is embedded during both the training and evaluation phases.

Our framework is implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU. We employ the AdamW optimizer [46] with a weight decay of 0.02. The choice of AdamW is motivated by its decoupled weight decay regularization, which correctly integrates weight decay with adaptive gradient updates, yielding substantially better generalization performance and training stability. We configure the learning rate with an initial value of 1×10^{-3} and systematically anneal it down to 1×10^{-6} via a cosine schedule [47]. This specific learning rate schedule allows the network to rapidly escape suboptimal local minima during the early stages of training while ensuring stable and fine-grained convergence in the final stages. The entire framework is trained for a total of 150 epochs. The model is trained until the composite loss stabilizes, ensuring that both image fidelity and watermark extraction performance have converged.

We evaluate the performance of our method through both digital simulations and SC experiments. By default, all experiments are conducted at an image resolution of 256×256 , unless explicitly stated otherwise. To ensure a fair comparison, all tests are evaluated on identical image subsets. Following the evaluation method of previous robust watermarking methods, we generate random secret messages for each image. An adequate quantity of samples can effectively mitigate the influence of fluctuations in the bit accuracy of the extracted messages, thereby ensuring consistent and reliable performance evaluation.

Digital Noise Evaluation: All digital noise analyses are conducted on a subset of 1,000 images randomly selected from the test set. This includes robustness tests against common digital noises.

Screen Camera Evaluation: For the SC experiments, a subset of 100 images is randomly chosen from the test set. To evaluate the robustness against real-world acquisition distortions, we capture these images from various viewing distances and angles. To ensure the stability and consistency required for ablation studies and comparisons with state-of-the-art methods, we established a controlled capture environment. In this setup, the camera was affixed to a tripod to maintain a stable position, and a Bluetooth remote shutter was used to eliminate motion blur from manual pressing. To mitigate the influence of fluctuating ambient light, comparative images were captured in immediate succession. By default, the capture was conducted at a distance of 30 cm with a frontal view. Our default hardware includes a Lenovo Legion Y9000P screen and a Samsung Galaxy S20 FE camera. To evaluate the generalization capability of our method, we also test on a broader range of devices: two additional screens (ASUS ROG Strix SCAR Edition 8, Envision G249G) and two additional cameras (iPhone 13, Vivo S17t). In our SC evaluation, we employ unconstrained devices rather than highly calibrated equipment. Because real-world piracy scenarios are inherently uncontrolled, the hardware characteristics utilized by adversaries are highly unpredictable. By capturing the test images displayed on screens using everyday smartphones, we objectively assess the framework's robustness against natural optical distortions. This evaluation protocol aligns directly with prior foundational works in the robust watermarking domain (e.g., StegaStamp [3], PIMoG [4], and SRWSS [11]), ensuring a fair and highly realistic assessment of SC robustness alongside existing baselines.

B. Metrics

Imperceptibility and robustness are key indicators of watermarking algorithms. The Peak Signal-to-Noise Ratio (PSNR) measures the difference between the watermarked and cover images, with higher values indicating less visual impact. Bit Accuracy (ACC) assesses the accuracy of the extracted message, with higher values indicating stronger robustness.

C. Baselines

We compare ReaguMark with several state of the art watermarking methods. For digital robustness, we include HiDDeN [17], MBRS [19], DWSF [43], and the resolution agnostic RAIMark [8]. For screen camera robustness, we compare with methods tailored for SC distortions: StegaStamp [3], PIMoG [4], and SRWSS [11]. Where available, we adopt the official implementations provided by the original authors to reproduce baselines for a fair comparison. The SC methods are adapted to arbitrary resolutions [7] following residual scaling techniques for fairness.

Due to the absence of publicly available source code for RAIMark, we evaluate its performance using the metrics reported in the original paper. This decision was made to ensure

fairness, as our initial local reproduction attempt yielded a noticeable performance gap due to undisclosed implementation specifics. In our quantitative tables, these results are explicitly marked with a dagger (†) to ensure transparency. While both frameworks utilize the COCO dataset, minor statistical variances may exist due to potential differences in the specific image testing subsets. Furthermore, because RAIMark focuses exclusively on digital domain distortions, our direct numerical comparison is confined to digital-domain metrics. Nevertheless, compared to RAIMark's reported results, ReaguMark not only achieves competitive visual quality but also successfully extends the resolution-agnostic watermarking capability to the much more challenging SC scenarios, which RAIMark does not address.

D. Comparative Experiments

1) Watermarking Performance against Digital Noise:

a) *Imperceptibility and Visual Quality:* The primary requirement for a watermarking scheme is that the embedded watermark should be imperceptible. As shown in Fig. 4, the watermarked images generated by our ReaguMark framework are almost indistinguishable from the cover images, maintaining high visual fidelity. In contrast, residual artifacts are noticeable in the images produced by other methods, especially HiDDeN and DWSF. Quantitatively, as listed in Table I, our method achieves a PSNR of 46.57 dB, which is over 6 dB higher than the best-performing baseline (RAIMark at 39.61 dB), confirming its superior imperceptibility.

b) *Attack Robustness:* We first evaluate the robustness against a wide range of common digital attacks. As presented in Table I, our ReaguMark framework demonstrates outstanding performance. It achieves perfect or near-perfect ACC across all attack types, including Gaussian Noise, Median Filtering, and Resizing. Even against challenging JPEG compression, it maintains an ACC of 99.26%. While RAIMark also shows strong digital robustness, our method achieves this state-of-the-art performance with a significantly higher level of imperceptibility, showcasing a trade-off between robustness and visual quality.

c) *Resolution-Agnostic Performance:* A core contribution of ReaguMark is its resolution-agnostic ability to handle arbitrary resolutions without retraining. We evaluate this capability in Table II. The results clearly show that our method maintains exceptional and stable performance across a wide spectrum of resolutions, from 256×256 to 1080×1920 . The PSNR remains consistently high (around 45.5-46.5 dB) and the average digital attack ACC is consistently above 99.89%. In contrast, MBRS is limited to a single resolution. Other methods, such as HiDDeN and DWSF, when applied to different scales, show fluctuating performance. Even compared to the resolution-agnostic RAIMark, our method provides a significantly higher PSNR while delivering comparable or superior ACC, demonstrating the effectiveness of our INR-based continuous representation and unified-scale embedding strategy.

2) Watermarking Performance against Screen-Camera:

TABLE II: Digital noise robustness test at different resolutions. We report PSNR (dB) / ACC (%). Our method maintains consistently high PSNR and ACC across all resolutions, unlike baselines. † indicates metrics directly cited from the original paper.

Models	Resolution (PSNR / ACC)					
	256×256	384×384	512×512	480×854	720×1280	1080×1920
HiDDeN	35.31 / 92.21	35.50 / 92.87	35.80 / 92.65	35.97 / 92.64	37.14 / 92.28	38.28 / 92.64
MBRS	27.63 / 92.94	- / -	- / -	- / -	- / -	- / -
DWSF	37.45 / 92.44	40.96 / 92.05	43.28 / 92.02	45.88 / 91.82	46.44 / 94.01	45.71 / 93.94
RAIMark [†]	39.61 / 99.88	39.83 / 99.90	39.79 / 99.96	39.73 / 99.79	39.73 / 99.90	39.73 / 99.92
ReaguMark	46.57 / 99.89	45.56 / 99.93	45.82 / 99.96	45.63 / 99.94	45.66 / 99.94	45.59 / 99.93

TABLE III: SC robustness comparison across different viewing angles and distances. We report ACC (%). All methods were calibrated to a PSNR of 45.5 dB for a fair comparison.

Models	Left			Right			Top			Bottom			Distance			
	60°	40°	20°	20°	40°	60°	60°	40°	20°	20°	40°	60°	20 cm	30 cm	40 cm	50 cm
StegaStamp	82.20	84.20	84.20	83.16	83.20	79.76	83.26	83.56	84.20	79.36	79.53	74.50	81.46	82.40	82.70	82.36
PIMoG	73.43	78.56	81.80	80.20	78.37	76.86	74.93	79.33	81.50	78.63	78.24	76.56	73.30	81.06	76.33	83.10
SRWSS	59.63	57.37	58.80	59.10	52.70	55.23	58.66	52.70	57.63	58.26	53.36	50.86	55.70	59.86	55.17	58.90
ReaguMark	92.57	95.76	96.44	96.47	95.80	94.04	96.37	96.20	96.60	95.80	95.67	90.03	93.44	97.10	94.67	97.13

TABLE IV: Comparison of watermark imperceptibility (PSNR in dB) under SC noise layer at default intensity.

Models	256×256	384×384	512×512	480×854	720×1280	1080×1920
StegaStamp	33.49	33.50	33.41	33.42	33.41	33.50
PIMoG	37.57	37.26	37.46	37.41	37.41	37.41
SRWSS	36.49	35.90	36.25	36.18	36.18	36.17
ReaguMark	46.09	45.36	45.34	45.53	45.50	45.54

TABLE V: SC ACC Comparison across Different Resolutions (%). All methods calibrated to PSNR $\approx$ 45.5 dB.

Models	256×256	384×384	512×512	480×854	720×1280	1080×1920
StegaStamp	82.40	84.06	84.50	86.26	84.20	82.90
PIMoG	81.06	83.63	84.53	85.80	86.86	80.93
SRWSS	59.86	65.26	61.26	66.73	67.23	59.60
ReaguMark	97.10	98.54	97.00	98.17	98.60	97.77

a) *Imperceptibility and Visual Quality*: To evaluate the quality of our proposed ReaguMark, we compare it with existing SC-oriented methods. First, with the default noise intensity factor of 1.0, our method produces a much higher PSNR (45.56 dB) compared to PIMoG (37.57 dB) and StegaStamp (33.49 dB), as shown in Table IV. This indicates that our data-guided noise simulation is inherently less destructive and more faithful to the original image content. Furthermore, as illustrated in Fig. 4, StegaStamp introduces severe visible artifacts, while SRWSS tends to generate high-frequency contour-like distortions. In comparison, ReaguMark and PIMoG exhibit similar artifact patterns, yet ReaguMark achieves better imperceptibility both quantitatively and perceptually.

b) *Attack Robustness*: To ensure a fair comparison that isolates robustness from embedding strength, we have calibrated all SC-resistant methods to produce watermarked images with a uniform PSNR of approximately 45.5 dB before conducting the tests by adapting the strength factor.

Robustness to Angles and Distances. Real-world SC captures tend to involve geometric distortions from varying

viewing perspectives. Table III presents the results of capturing watermarked images from different angles and distances. Our method demonstrates significant improvement, maintaining an ACC of over 90% even at extreme 60° angles, and consistently high ACC (93%-97%) at different distances. This confirms our framework’s effectiveness against the perspective and resolution changes common in practical SC scenarios.

Cross Device Generalization and Unseen Distributions.

To evaluate cross-device generalization, we analyze how varying SC degradations introduced by different camera-screen combinations impact the extraction robustness. In practice, discrepancies in screens and cameras inherently produce unpredictable moiré patterns and illumination variations. To mitigate performance degradation under such hardware discrepancies, our LAIN module statistically aligns luminance distributions rather than memorizing fixed color profiles. Furthermore, by leveraging a diverse distribution of real-world moiré captures, the RMPS module encourages the network to learn more generalized features for moiré patterns.

To evaluate this generalization capability, we test the models across nine distinct pairs of screens and cameras. Notably, the moiré patterns utilized during our training phase are sourced exclusively from the external UniDemoiré dataset, ensuring no hardware overlap with our testing combinations. Consequently, this cross-device evaluation assesses the model under strictly unseen physical degradations. As shown in Table VII, ReaguMark consistently demonstrates the highest robustness against diverse hardware variations, achieving extraction accuracies frequently exceeding 97% and significantly outperforming all baselines. These results validate that our data-driven noise layer effectively models a generalized distribution of SC distortions, enabling robust message extraction across unseen hardware environments.

c) *Resolution-Agnostic Performance*: As shown in Table V, our method outperforms all baselines across various resolutions in SC. ReaguMark consistently achieves an ACC above 97%, while the next best method, PIMoG, struggles

TABLE VI: Combined ablation study for intermediate scale (Resolution) and training-time scale factors (Scale Range).

Type	Setting	AVG PSNR (dB)	Identity	GN(0.05)	MF(7)	JPEG(50)	Crop(0.25)	Resize(0.5)	Resize(2.0)	SC	AVG BER
Resolution	112×112	46.21	100.00	100.00	100.00	99.68	88.67	100.00	100.00	95.14	97.94
	128×128	46.50	100.00	100.00	100.00	99.26	99.95	100.00	100.00	97.10	99.54
	144×144	46.76	100.00	100.00	100.00	97.80	96.9	100.00	100.00	97.13	98.98
Scale Range	[1, 1]	48.15	99.76	50.00	95.25	98.60	99.83	50.00	99.60	81.60	84.33
	[1, 2]	46.50	100.00	100.00	100.00	99.26	99.95	100.00	100.00	97.10	99.54
	[1, 3]	47.83	100.00	100.00	100.00	95.34	99.90	100	100	97.43	99.08

TABLE VII: Cross-device SC robustness ACC (%). All methods calibrated to PSNR $\approx$ 45.5 dB.

Cameras	Methods	Screens		
		Lenovo	ASUS	Envision
Samsung	StegaStamp	82.40	84.35	85.00
	PIMoG	81.06	82.63	83.54
	SRWSS	59.86	55.76	58.91
	ReaguMark	97.10	97.88	98.84
iPhone	StegaStamp	83.92	84.76	84.61
	PIMoG	82.51	85.70	83.30
	SRWSS	59.96	60.76	60.43
	ReaguMark	97.60	98.11	99.00
Vivo	StegaStamp	83.12	83.67	84.10
	PIMoG	76.23	79.63	78.91
	SRWSS	56.73	61.56	59.44
	ReaguMark	93.32	94.75	97.07

TABLE VIII: Comparison of different Screen Camera noise layers.

Noise Layer paradigm	PSNR (dB)	ACC (%)
StegaStamp	42.34	92.72
PIMoG	45.56	95.83
ReaguMark	46.09	97.10

to exceed 86%. SRWSS, despite being data-driven, shows poor performance, likely due to its fixed-resolution. In terms of imperceptibility, Table IV further shows that ReaguMark maintains uniformly high PSNR across all resolutions at default intensity, staying around 45.5 dB with less than 0.8 dB fluctuation. In comparison, StegaStamp, PIMoG, and SRWSS exhibit considerably lower PSNR, with ReaguMark outperforming them by 8–12 dB across all tested resolutions. These results highlight the combined strength of our resolution-agnostic design: ReaguMark delivers both cross-scale robustness and cross-scale imperceptibility, without retraining or reconfiguration.

E. Ablation Studies

We structure our ablation studies into two distinct evaluation scopes. For network hyperparameter ablations, we evaluate performance across both comprehensive digital distortions and SC degradation. The ablation of our proposed SC-specific noise layers (LAIN and RMPS) is conducted exclusively under SC scenarios. Although the SC noise layer can be used for digital noise, its core design serves SC scenarios. Therefore, its performance is most meaningful when evaluated against actual physical distortions. This approach ensures that we assess each

TABLE IX: Internal module ablation of the ReaguMark noise layer under Screen Camera conditions.

Proposed Modules		Metrics	
LAIN	RMPS	PSNR (dB)	ACC (%)
-	-	45.44	94.56
✓	-	45.86	96.47
-	✓	45.85	96.37
✓	✓	46.09	97.10

TABLE X: Single Variable Sensitivity Analysis of the Illumination Perturbation Factors in the LAIN module.

Fixed Parameter	Varying Parameter	Value	PSNR (dB)	ACC (%)
Std Range ($\epsilon_\sigma = \pm 0.05$)	Mean Range (ϵ_μ)	± 0.00	45.82	96.55
		± 0.05	45.96	96.52
		± 0.10	46.09	97.10
		± 0.20	45.94	96.85
Mean Range ($\epsilon_\mu = \pm 0.10$)	Std Range (ϵ_σ)	± 0.000	45.78	96.41
		± 0.025	46.02	96.55
		± 0.050	46.09	97.10
		± 0.100	46.05	96.30

component's contribution specifically under the conditions it was engineered to solve.

1) *Effectiveness of Noise Layer Components*: We first validate the specific contributions of our proposed SC noise layer. We evaluate the performance under SC conditions by separating the evaluations into two distinct analyses: a comparison of different noise layers (Table VIII) and an internal module ablation of our framework (Table IX).

As observed in Table VIII, replacing our entire noise layer with those from StegaStamp or PIMoG leads to a drop in ACC, proving the effectiveness of our data guided simulation strategy. Furthermore, Table IX demonstrates that employing the complete noise layer equipped with both LAIN and RMPS achieves the SC robustness.

We attribute the performance degradation upon module removal to the unmitigated physical complexities of SC simulation. When the LAIN module is removed, the framework struggles to adapt to the complex global color casting, ambient brightness variations, and contrast shifts inherent in physical captures. Consequently, the decoder becomes more sensitive to illumination deviations, leading to a decrease in ACC. Conversely, when the RMPS module is removed, the training pipeline lacks exposure to moiré patterns. As a result, the decoder becomes more vulnerable to these patterns, which results in a reduction in ACC.

2) *Effect of Unified Intermediate Scale*: Table VI investigates the choice of the unified intermediate resolution. We find

TABLE XI: Single Variable Sensitivity Analysis of the Moiré Synthesis Weights in the RMPS module.

Fixed Parameter	Varying Parameter	Value	PSNR (dB)	ACC (%)
Grain Merge ($w_{GM} = 0.100$)	Multiply Weight (w_{MU})	0.000	45.85	96.65
		0.0625	46.01	96.35
		0.125	46.09	97.10
		0.250	45.92	96.05
Multiply Weight ($w_{MU} = 0.125$)	Grain Merge (w_{GM})	0.000	45.72	96.62
		0.050	45.95	96.50
		0.100	46.09	97.10
		0.200	45.98	96.25

TABLE XII: Statistical analysis of robustness. We report the Mean $\pm$ Std alongside the 95% CI of the ACC (%) computed across the tests.

Evaluation Scenario	Mean $\pm$ Std (%)	95% CI
Cross Resolution Variations		
Resolution 256 $\times$ 256	97.10 $\pm$ 5.70	[95.97, 98.23]
Resolution 512 $\times$ 512	97.00 $\pm$ 6.79	[95.65, 98.35]
Resolution 1080 $\times$ 1920	97.77 $\pm$ 5.63	[96.65, 98.88]
Cross Device Variations		
Lenovo Legion & Samsung S20	97.10 $\pm$ 5.70	[95.97, 98.23]
ASUS ROG & iPhone 13	98.11 $\pm$ 4.85	[97.15, 99.07]
Envision Monitor & Vivo S17t	97.07 $\pm$ 6.12	[95.86, 98.28]

that a scale of 128×128 provides the best overall performance, achieving a high average robustness of 99.54%. A smaller scale shows a clear drop in robustness against cropping, while a larger scale slightly degrades performance against JPEG compression. This validates our choice of 128×128 as an effective trade-off for feature representation. The choice of the unified intermediate scale is flexible and can be manually specified. Other values are also feasible, and the framework does not depend on a fixed scale.

3) *Effect of Training-time Scale Perturbation*: Finally, we explore the impact of the random scale range used during training in Table VI. Training on a fixed scale ($[1, 1]$) fails completely against Gaussian noise and resize attack (50% ACC) and performs poorly in SC (81.60% ACC). Introducing a scale range of $[1, 2]$ dramatically improves performance across the board, achieving an average of 99.54%. Further widening the range to $[1, 3]$ does not yield additional benefits and shows a drop, suggesting that the $[1, 2]$ range is sufficient and optimal for learning scale invariance.

4) *Parameter Sensitivity Analysis*: To investigate the sensitivity of the noise layer modules to their hyperparameter settings, we conduct a control variable analysis. We independently examine the illumination perturbations in the LAIN module and the blending weights in the RMPS module.

a) *Illumination Perturbation Factors*: In the LAIN module, we apply random uniform perturbations to the channel mean μ and standard deviation σ , varying one parameter while keeping the other fixed. The results are summarized in Table X. The ACC shows a consistent trend across the evaluated ranges, reaching 97.10% with a PSNR of 46.09 dB at $\epsilon_\mu = \pm 0.10$ and $\epsilon_\sigma = \pm 0.05$. Both smaller and larger perturbation ranges tend to yield lower performance, suggesting that these values represent an effective hyperparameter configuration.

b) *Moiré Synthesis Weights*: For the RMPS module, we independently examine the synthesis weights associated with the multiply (w_{MU}) and grain merge (w_{GM}) operations. The evaluation is presented in Table XI. The results indicate a consistent trend around $w_{MU} = 0.125$ and $w_{GM} = 0.100$, where the ACC reaches 97.10%. The observations suggest that these specific intervals provide a stable range, making this configuration a reasonable choice for our subsequent evaluations.

F. Statistical Analysis of Robustness

To further evaluate the performance of ReaguMark, we conduct a statistical variance analysis across our testing samples. While average ACC provides a general performance overview, it may not fully capture potential instabilities on challenging individual samples. Therefore, we calculate the Standard Deviation (Std) and 95% Confidence Interval (CI) of the ACC across diverse SC test scenarios. Since each captured image represents an independent physical instance, this sample-wise evaluation helps quantify the performance consistency of our framework under varying conditions. The statistical results, encompassing diverse cross-device combinations and resolutions, are summarized in Table XII. The statistical results indicate that ReaguMark maintains a stable accuracy. Notably, with the lower bounds of the 95% confidence intervals remaining above 95.6% across all tests, the framework demonstrates reliable performance.

V. CONCLUSION

In this paper, we propose ReaguMark, a Resolution-Agnostic Unified Watermarking framework that addresses the challenges of resolution variance and SC distortions. Our method adopts a resolution-agnostic training strategy with INR for seamless watermark embedding and extraction. We further design an SC noise layer guided by real data, incorporating LAIN and RMPS. Extensive experiments, including ablation studies, demonstrate that ReaguMark surpasses state-of-the-art methods in robustness, imperceptibility, and especially cross-scale consistency under both digital and SC conditions. Future work will focus on extending the framework to handle more complex distortions and exploring self-adaptive watermarking mechanisms for real-time SC scenarios.

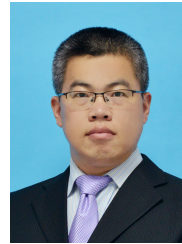
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